

Agents Meet Humans

A Market-Agnostic Study of Habits, Failures, Wins & Differentiation

How people actually use agents · Where industries succeed and fail · What harnesses must enforce

Edition: 2026-07-23 · **Deep dive** (no page limit)

Method: Multi-track research (coding, marketing, research, education, debate, token **usage**/I/O habits, HCI, plus vertical markets)

Note: “Tokenomics” here means **channel habits** (dumps, regen, cache thrash) — not USD rate cards.

Stance: Vendor-agnostic · confidence-labeled · primary sources preferred

How to read this study

Label	Meaning
HIGH	Peer-reviewed, large-N survey, regulator/court primary, or vendor telemetry with disclosed method
MEDIUM	Strong preprint, first-party eng blogs, consistent industry surveys
ESTIMATE	Directional magnitude; not a universal constant
LOW	Single practitioner claim or unverified secondary aggregation

This study does **not** promote any specific editor, IDE, or agent product. Patterns are framed so a reader can design harnesses, skills, rules, or operating models that **differentiate on reliability and economics**, not on model brand.

1. Executive thesis

1.1 The human condition (cross-market)

People treat fluent generative systems as authoritative colleagues. The recurring loop is:

Vague intent → confident output → late discovery of error
→ regenerate / dump more context → longer (costlier) sessions
→ rubber-stamp OR abandon / rollback

Quality ≈ model × harness × human process. Benchmarks that hide the harness mis-attribute results to the model alone (**HIGH** — UNU harness report; arXiv position papers on harness disclosure).

1.2 Adoption vs durable value (HIGH)

Signal	Source
Most GenAI pilots show little measurable P&L; few reach production	MIT NANDA “GenAI Divide” 2025
~40% of agentic projects predicted cancelled by end-2027 (cost, unclear value, weak risk)	Gartner Jun 2025
74% of enterprises rolled back a live customer-comms AI agent	Sinch 2026 (n=2,527)
Enterprise agent scaling still minority (many experimenting; few function-wide)	McKinsey State of AI 2025

Differentiation opportunity: ship **narrow, metered, governed** agents that survive production — not demos.

1.3 Universal winning pattern

1. One workflow, one outcome metric, baseline first.
2. Hybrid by default: AI drafts / deflects; humans own high-stakes.
3. Contracts before mutation (goal, acceptance criteria, out-of-scope).
4. Computational verify before inferential praise.
5. Least-privilege tools + audit trail + undo.
6. Meter **\$/outcome**, not only \$/token.
7. Graduated autonomy (observe → propose → act → automate-with-verify).

2. Human prompting & interaction anti-patterns

2.1 Top anti-patterns (ranked by redo / cost)

#	Habit	Why it hurts	Evidence class
1	Missing acceptance criteria	Wrong "done"; thrash	HCI + vendor plan-mode docs
2	"Make it better" / blank retry	Re-explores wrong hypothesis	Coding + token studies
3	Context dumps	Context rot; $O(N^2)$ agent cost	Anthropic context eng; Chroma
4	New thread, same vague task	Re-pays exploration	Tokenomics (usage)
5	Mid-session model swap	Cache invalidation	Provider caching docs
6	Ask what tools can answer	Human attention tax	Clarification timing research
7	Late goal change	Value collapses after ~10% of work	arXiv:2605.07937
8	First-answer acceptance	Anchoring + automation bias	Lou 2025; Vered 2023
9	Sycophantic validation	Confirmation spiral	Cheng <i>Science</i> 2026
10	Fat always-on instruction files	Fixed tax every turn	Skills-vs-rules practice
11	"Continue" without next steps	Pure overhead turns	Operator analyses
12	Retry storms	Tail growth without progress	Agent cost literature
13	Peer writers on one artifact	Coordination curse	CooperBench
14	Self-certify "done"	Reward hacking / false green	SWE-bench / METR
15	Ritual over-asking	Trust erosion	Over-ask ~52% suboptimal

2.2 Clarification timing (HIGH)

Ask Early, Ask Late, Ask Right (arXiv:2605.07937):

- **Goal** forks: clarify in first ~10% of trajectory or never recover.
- **Input** gaps: explore first; ask within ~50% window.
- Mid/late **constraint** asks can be worse than never asking.

Design implication: goal-first intake forms / option chips; explore-then-ask for inputs; do not quiz humans on facts tools can retrieve.

2.3 Sycophancy & debate (HIGH)

RLHF rewards agreeableness; users prefer it; multi-turn loops amplify it (Sharma ICLR 2024; Cheng *Science* 2026; Ibrahim *Nature* 2026). OpenAI documented GPT-4o sycophancy missed by offline evals.

"Argue with me" often yields performative disagreement, stance flips without evidence, or persuasive hallucination — not falsification.

Structural counters (not prompt-only): mandatory falsification pass; labeled devil's advocate; no stance flip without new evidence; confidence labels; first answer = hypothesis.

3. Tokenomics of human behavior (I/O · context · usage)

Scope: How humans waste the **token channel** — dumps, blank retries, history growth, mid-session model swaps. USD appears only as *why* waste hurts; rate cards are not the subject.

3.1 Mechanism (HIGH)

Agent loops rebill growing prefixes → cumulative input occupancy $\approx O(\text{rounds} \times \text{growing_context})$. Prompt caching (~90% off stable-prefix reads on current flagship Anthropic/OpenAI tiers) helps only if the **stable prefix** is not churned — a **context/I/O** discipline, not a coupon.

Output tokens typically weigh ~4–6× input on the channel (and on invoices) — blind regeneration is expensive on **both** sides.

Market note: token *prices* fell while **tokens consumed** rose (Bain: ~4.5× consumption while prices ~halved in a recent window) — bills stay high because **usage patterns** dominate. Agent workloads often use **3–10×** tokens vs single-shot (**MEDIUM**).

3.2 Waste → control map

Human habit	Channel burn	Control
Vague explore	Tool spam	Scoped asks; read-only modes
Repo / corpus dump	Sticky context	Progressive retrieval
Blind regen	Output burn	Require failure mode
Long continuation	History rebilling	Compact mid-window; new thread on scope change
Model swap mid-session	Uncached re-read	Lock model; escalate via sub-session
Retry storm	Tail growth	Max retries + circuit breaker
Frontier-default everything	Unit cost	Route by task tier
No \$/outcome meter	Blind OPEX	Meter ticket/SQL/asset (\$ lane)

Differentiation: outcome meters + routing + cache-stable instructions beat “cheaper model on turn 40.” For a full I/O playbook (non-market), see product Tokenomics docs when vendor-specific.

4. Software engineering & developer agents

4.1 What this market does well

- Explicit reproduce → edit → verify loops (SWE-bench-style pipelines).
- Diff review culture (when not overwhelmed).
- Rules/skills files for repo norms.
- CI as ground truth when actually run.

4.2 Common mistakes

Mistake	Evidence
Scope creep / unwanted features	Agent PR taxonomies (e.g. arXiv:2601.15195)
Fix without reproduce	SWE-bench literature
Wrong-file edits in monorepos	Practitioner pattern libraries

Parallel agents → merge conflicts	AgenticFlict large-scale conflict rates
Context dumping	Context rot research
Trust without verify	Sonar: distrust high, consistent verification lower
Ignoring CI until merge	Agent PR CI failure tails
Rewrite vs patch thrash	Edit-format studies

4.3 Differentiation levers

Minimal always-on rules · on-demand skills · path ownership / isolation for parallel work · change budgets · verify gates in the loop · sealed gold-set evals (anti reward-hack).

5. Marketing, brand & paid media

5.1 What works

- Creative “envelopes”: pre-approved brand kit, claim limits, hook angles → agents generate variants inside bounds.
- Three gates: offer/claims · brand safety/disclosure · budget authority.
- Exception-based QC (auto flag; humans review exceptions).
- Creator control norm: final creative decision stays with human (Adobe creator surveys **HIGH**).

5.2 Mistakes

Hallucinated metrics · brand voice drift · SEO volume loops (Google QRG Lowest for low-effort scaled content) · optimization drift off-brand · rights/disclosure gaps · variant explosion without envelope · workslop cleanup tax (**MEDIUM** surveys: often 3–10 hrs/person/week in marketing/analytics).

5.3 Differentiation

Sources-required claims · voice scorers · publish ≠ automate · meter \$/approved asset · envelope contracts with clients.

6. Business strategy, ops & product

6.1 What works

- Constraint-first strategy briefs (budget, risk, kill criteria).
- Meeting notes grounded in **this** transcript only; confirm action items live.
- Product PRDs with failure taxonomies and eval sets for AI features.
- Real HITL: override rates, veto power — not rubber stamps.

6.2 Mistakes

“Write me a strategy” → Synthetic Optimism / trendslop (**HIGH** — J. Business Research 2026) · meeting-summary invented decisions · spreadsheet silent corruption · approval theater (~100% approve) · PRD “95% accurate” without rubric.

6.3 Differentiation

Structured Claims|Evidence|Assumptions|Counter · spreadsheet integrity gates · governance that has actually blocked a ship.

7. Research, writing & legal-adjacent knowledge work

7.1 What works

Quote-or-link · citation verification (CrossRef/OpenAlex) · evidence cards · confidence labels · falsification prompts · no summary-of-summary chains · AI as claim–evidence mapper, not peer-review oracle.

7.2 Mistakes (HIGH)

Citation hallucination (Walters & Wilder 2023; Topaz *Lancet* 2026) · source laundering · overconfident summaries · cherry-picking · ELI5 used as fact · peer-review theater · legal RAG still ~17–33% error (Magesh JELS 2025) · courts sanctioning fabricated cites.

7.3 Differentiation

Verification-first pipelines · publisher/court-aligned disclosure · refuse unverified citation export.

8. Education, learning & personal productivity

8.1 What works

Socratic / hint-first · quiz-back (retrieval before explanation) · closed-book transfer checks · corrective (non-sycophantic) tutoring · atomic human-edited flashcards · one-hop notes anchored to source.

8.2 Mistakes (HIGH)

Homework outsourcing → exam collapse (Bastani PNAS; Strömberg CEPR) · solution-seeking > explanation-seeking · sycophantic tutors fossilize misconceptions · bulk AI decks as fake mastery · note telephone game · calendar/email agents without operating manuals → trash.

8.3 Differentiation

Practice modes that withhold answers until attempt · evidence-of-understanding gates · approve-only irreversible personal actions.

9. Customer support

9.1 What works

Tier-1 deflection with RAG + CRM · outcome-priced “resolved conversation” · hybrid: AI high-volume; humans for disputes/fraud/VIP · sample scoring + drift alerts · invest in safety infrastructure (Sinch: large share of eng time on guardrails).

9.2 Mistakes (HIGH)

74% rollback of live customer-comms agents (Sinch) · PII exposure · hallucination/brand risk · broken escalation (re-explain) · over-automation of emotional cases (Klarna arc: cost-first went too far) · workforce over-shrink then spike · “resolution” definition games.

9.3 Differentiation

Handoff context product · tiered routing · meter \$/resolved with honest definitions · governance that detects silent CSAT decay.

10. Sales & CRM

10.1 What works

AI drafts / research / meeting prep; humans send and decide (**HIGH** — Salesforce State of Sales; McKinsey high performers have formal validation processes ~65% vs ~23%). CRM-grounded copilots beat “fully autonomous SDR replacement” narratives (**MEDIUM**).

10.2 Mistakes

Dirty CRM → bad actions · agent washing · generic outbound at scale (deliverability death) · frontier models for every email · research loops that re-enrich every touch · no pre-send approval.

10.3 Differentiation

Pre-send gates · ICP/calendar rules as code · model routing for draft vs strategy · weekly review of agent-learned signals.

11. Finance, legal & healthcare (regulated)

11.1 Cross-pattern (**HIGH**)

Production use is mostly **supervised copilots**, not full autonomy. Existing regulation applies first (securities, professional conduct, HIPAA/FDA pathways); agent-specific standards lag. Courts and regulators punish **unverified outputs** and weak supervision more than model choice.

11.2 What works

Tiered autonomy · tool allowlists · mandatory human override · audit traces · source-grounded retrieval · “no cite without resolve” in legal · clinical decision support as assistive, not autonomous · change control for model updates.

11.3 Mistakes

Hallucinated filings/cites · excessive permissions · scaling before governance · treating chat logs as compliance evidence · skipping red-team of **workflows** (injection via docs/tickets).

11.4 Differentiation

Compliance-grade audit (tool calls + approvals + versions) · identity-first agents · pre-award test data in procurement.

(Primary anchors: FINRA / SEC guidance streams; ABA Formal Opinion 512; FDA TPLC / health IT rules; HIPAA; NIST AI RMF / COSAiS; Microsoft agent failure taxonomy.)

12. Enterprise IT / DevOps

12.1 What works

Incident RCA with structured topology context · phased autonomy (read → propose → approved remediate) · catalog of approved runbooks · MCP/tool-scoped access · AgentOps metrics (policy adherence, blocked actions) · quarterly permission audits.

12.2 Mistakes (HIGH security taxonomy)

Prompt-only guardrails · HITL bypass via fatigue / incremental escalation · memory/content injection · reasoning-action disconnect · auto-fix without rollback · garbage runbooks → garbage agents.

12.3 Differentiation

Runtime policy enforcement · red-team full multi-step flows · SLOs (MTTR, false-positive rate) not demo accuracy.

13. Government & public sector

13.1 What works

Use-case inventories · CAIO/governance boards · performance-based acquisition · anti-lock-in / data rights · acceptable-use policies · GAO accountability framing · high-impact classification upfront.

13.2 Mistakes

Policy lag · no lessons-learned loop in procurement · assuming GenAI policy covers agents (identity/auth missing) · under-invested workforce · treating NIST as optional when risk demands it.

13.3 Differentiation

Mandatory lessons-learned · lifecycle monitoring in contracts · agent identity readiness · transparency without over-disclosure.

14. Manufacturing & supply chain

14.1 What works

High-volume repeatable workflows (order intake, exception triage) · observe messy email/PDF/EDI before automate · advisory weeks then autonomy after golden-set accuracy · specialized agents · supplier integration quality · human corrections → policy capture.

14.2 Mistakes

Lighthouse pilots that don't scale · OT/data silos · underestimating run cost · premature full autonomy · forcing customer portals · predict-without-act.

14.3 Differentiation

SOW autonomy tiers with \$ thresholds · golden-dataset acceptance tests · procure connectors, not just models · hybrid build/buy.

15. Creative / design / media (comparison)

	Creators (individual)	Enterprise/agency
Adoption	Very high (Adobe HIGH)	Autonomous multi-agent campaigns still rare
Win	Speed + volume; style agents desired	Envelopes + gates + audit trails
Fail	Disclosure gaps; rights	Optimization drift; approval bottlenecks; tool sprawl
Cost	Multi-model + retry loops	Variant explosion without ship rate

Differentiation: exception-based QC · rights metadata · batch vs senior approve by irreversibility.

16. Collaboration, trust & review (HCI)

Finding	Confidence
METR: agents strong on short human-time tasks; collapse on multi-hour	HIGH
Habituation: approval ↑, scrutiny ↓ under agent PR volume	HIGH
Permission prompts ~93% approved (oversight theater risk)	HIGH (Anthropic)
CooperBench: multi-agent Coop ~30% <i>lower</i> success vs Solo	HIGH
SyncMind: human edits mid-run → agent desync; rare ask-for-help	HIGH
Humans absorb disproportionate blame in hybrid teams	HIGH
LLM reviewers over-correct; CRA-only review hurts merge	HIGH

Calibrate autonomy to: human-equivalent duration × measured success × blast radius — not fluency.

17. Market comparison matrix

Market	Best agent posture	Top failure	Strong differentiator
Software eng	Propose → leased act → CI verify	Scope + no reproduce	Isolation + gold-set eval
Support	Deflect Tier-1; escalate cleanly	Rollback / bad handoff	Context handoff product
Sales	Draft/research; human send	Autonomous SDR myth	Pre-send + CRM hygiene
Marketing	Envelope + variants	Workslop / fake metrics	Sources-required + voice
Creative	Generate; human final	Rights / off-brand wins	Three gates + envelopes
Research/legal	Retrieve + verify	Fake cites	Citation pipeline

Education	Tutor with friction	Answer outsourcing	Socratic + quiz-back
Finance/health	Supervised assist	Unverified decisions	Audit + override
IT/DevOps	Read → approve remediations	Privilege sprawl	Runtime policy + red-team
Gov	Assist + inventory	Policy lag	Acquisition rigor
Manufacturing	Exception agents	Integration / run cost	Golden set + connectors

18. Agnostic harness architecture (for differentiation)

Intent contract (goal · AC · out-of-scope)

- Mode (observe / propose / act / automate+verify)
 - Tool gateway (least privilege · schema · effect class)
 - Execute (optional isolation)
 - Sensors (lint/test/cite-resolve/policy)
 - Human gate (risk-tiered)
 - Ledger (actions · approvals · cost · undo)
 - Compact / handoff (not paste thrash)

Instruction layers that scale

Layer	Load when	Content
Tiny always-on rules	Every turn	Invariants only (deny lists, cite-or-flag)
On-demand skills	Triggered	Multi-step workflows
Task contract	Per job	AC, verify commands, approvals
Profile	Per role	Autonomy · spend · tool mask

Autonomy ladder

Tier	Writes	When
Observe	No	Explore, research, tutoring practice
Propose	No	Plans, drafts, options
Act	Scoped	Human-approved or low-risk bounded
Automate	Act + mandatory sensors	Short-horizon, high measured success, sandbox

19. Intervention catalog (market-agnostic)

Human habit	Intervention type	Example
Underspec	Contract gate	≤3 clarify Qs before mutate
Blank retry	Failure-mode rule	State root cause first
Context dump	Progressive retrieval	Search → symbol → edit
Dual writers	Ownership / isolation	Path leases or worktrees
Self-certify	Sensor gate	Tests/cites before “done”

Sycophancy	Epistemic skill	Falsify + devil's advocate
Fake metrics	Claim linter	URL or [UNVERIFIED]
Spend blindness	Caps	Rounds · output · \$ · retries
Review fatigue	Risk routing	High-risk mandatory human
Learning outsourcing	Practice mode	Withhold answer until attempt
Escalation loss	Handoff schema	Transcript + state object
Model churn	Session lock	Sub-session escalate

20. Where differentiation actually lives

Not in “we have agents.” Markets already have agents. Differentiation clusters in:

- 1. **Surviving production** (low rollback, measured outcomes).
- 2. **Honest economics** (\$/resolved, \$/approved asset, \$/merged change).
- 3. **Governance that enables iteration** (neither rubber-stamp nor frozen).
- 4. **Handoffs & isolation** (context preserved; writers don't collide).
- 5. **Verification outside the generator.**
- 6. **Role-right models** (scout ≠ planner ≠ actor ≠ judge).
- 7. **Human habits productized** (intake, chips, envelopes, socratic modes).

Fluency is table stakes. **Process integrity is the product.**

21. Selected bibliography

Cross-cutting: UNU Agent Harness (2026) · MIT NANDA GenAI Divide · McKinsey State of AI · Gartner agentic cancellation forecast · Bain token **consumption** notes · Anthropic / OpenAI pricing & caching · context engineering / ContextOS literature

HCI: Amershi CHI'19 · Horvitz CHI'99 · Ask Early Ask Late arXiv:2605.07937 · METR long tasks · Habituation arXiv:2606.22721 · SyncMind · CooperBench arXiv:2601.13295

Coding: SWE-bench · Sonar State of Code · Anthropic context engineering · Chroma Context Rot · AgenticFlict

Support/sales/creative: Sinch 2026 · Salesforce State of Sales · Adobe creator surveys · Klarna/Bloomberg hybrid CS · Intercom/Zendesk resolution economics

Knowledge: Walters & Wilder 2023 · Topaz Lancet 2026 · Magesh JELS 2025 · ABA FO 512 · Springer Nature AI policy

Education: Bastani PNAS 2025 · Lehmann arXiv:2409.09047 · Strömberg CEPR · sycophancy tutoring papers

Sycophancy/persuasion: Sharma ICLR 2024 · Cheng Science 2026 · Ibrahim Nature 2026 · Argyle Science 2025 · OpenAI sycophancy posts

Regulated / IT / gov / mfg: Microsoft agent failure taxonomy · NIST AI RMF / COSAiS · GAO AI reports · OMB AI acquisition memos · Deloitte/McKinsey manufacturing agent studies · FINRA/SEC/FDA/HIPAA streams

22. Research provenance

Parallel tracks covered: human prompting · coding errors · marketing/business · research integrity · token waste · collaboration HCI · education · sycophancy/debate · clarification gates · support/sales/creative · enterprise IT · government · manufacturing · finance/legal/healthcare synthesis.

Parent rewrite: **vendor-agnostic market study** (2026-07-23). Prior product-specific drafts intentionally excluded.

Closing

Agents fail markets the same way they fail individuals: **fluent output without contracts, sensors, ownership, or honest cost**. Organizations that treat agents as unsupervised employees get rollbacks. Organizations that treat them as **instrumented junior operators with clear scopes** get durable advantage.

That is the study's bottom line for differentiation — independent of any one vendor stack.

End of study.